

THE PARTICLE THEORY OF MATTER

It is generally accepted by scientists that all matter is made up of particles. The following assumptions about matter apply.

1. All matter is made up of tiny particles.
2. All the particles in a substance are the same. Different substances are made of different particles.
3. There are attractive forces among particles. These attractions may be strong or weak.
4. The particles are always moving. The more energy the particles gain, the faster they move.
5. There are spaces among the particles.

STATES OF MATTER

SOLIDS

- Particles vibrate in a fixed position.
- Definite shape and volume.
- Strong forces between particles hold them in position.

LIQUIDS

- Particles are further apart and able to move past each other.
- Take the shape of the lowest level of a container.
- Definite volume, changing shape.

GASES

- Particles are very far apart and move freely in all directions.
- Very weak forces between particles.
- Fill the whole container.
- Changing shape and volume.

